UNIT	MANUFACTURING
SUBJECT	PROCESS FLOW

ANNEX A

Specifications and Limitations for PCB Fabrication and PCB Assembly

ITEM	DESCRIPTION	LIMITATIONS	
PCB Fabrication			
1	PCB Allowable Design Dimension (x,y)	190 mm x 250 mm Maximum	
2	PCB Material	FR4 only, client can provide own PCB	
3	Board Thickness	1.6mm only	
4	Copper Thickness	35µm (1oz)	
5	Solder Mask Color	Green, Black, Blue, Red, Orange, White, Yellow, Pink, Purple (subject to color availability)	
6	Silkscreen Color	Black and White (subject to color availability)	
7	Silkscreen Line Width	0.15 mm Minimum	
8	Through-Hole Plating Process	Chemical-Free, organic Conductive Polymer Paste (not recommended for high current application)	
9	Hole Diameter	Ø0.4 mm (15mil) Minimum	
10	Slotted Hole	1 mm Minimum	
11	Annular Ring	0.254 mm (10mil) Minimum	
12	Isolation type	a. Basic Isolation b. Rubout (incur additional cost)	
13	Pattern Width	0.3 mm (12mils) Minimum	
14	Pattern Space	0.254 mm Min. for 35µm copper	
15	Pad Surface Finishing	Bare Copper, Immersion Tin	
16	PCB design format	In Gerber File (RS 274-x)	
17	Number of Layers	1, 2	
18	Lead-time (Days)	To be determined based on the client's design	
19	Outside route to copper track	0.3mm (12mils) board outline, edge	
20	Drill Sizes	Spiral Drill 0.4mm (Green) Spiral Drill 0.5mm (Green) Spiral Drill 0.6mm (Green) Spiral Drill 0.7mm (Green) Spiral Drill 0.8mm (Green) Spiral Drill 0.9mm (Green) Spiral Drill 1.0mm (Green)	Spiral Drill 1.2mm (Green) Spiral Drill 1.4mm (Green) Spiral Drill 1.5mm (Green) Spiral Drill 1.6mm (Green) Spiral Drill 1.8mm (Green) Spiral Drill 2.0mm (Green)
We convert the drill size given in the drill files or tool lists into millimeter and round to the nearest 0.1mm. **Example**: Drill size of 31 mil – converted to 0.7874 mm is then rounded to 0.80 mm **Example**: Drill size of 32 mil – converted to 0.8128 mm is then rounded to 0.80 mm **Example**: Drill size of 33.5 mil – converted to 0.8509 mm. Will be rounded to 0.9mm if within minimum annular ring size.			

PCB Assembly		
1	Parts Acquisition	Service available in Consignment
2	Stencil (for SMD Components)	Material: Polyimide Max. Component Area: 210mm x 250mm
3	Surface Mount Components	0201 Min. Chip Package All kinds except Grid array types
4	Through-Hole Components	Radial and Axial
5	Standard (Reference)	IPC-A-610, Acceptability of Electronics Assemblies
6	Solder and flux	Lead-Free